

The cost of capital has two aspects. It is the cost of funds that a company raises and uses and the return that investors expect to be paid for putting funds into the company. It is the minimum return that a company should make on its own investments to earn the cash flow out of which investors can be paid their return.

The cost of capital can therefore be measured by studying the returns required by investors, and used to derive a discount rate for DCF analysis and investment appraisal.

Cost of capital, however it is measured, is an opportunity cost of finance, because it is the minimum return that investors require. If they do not get this return, they will transfer some or all of their investment somewhere else.

Here are two examples

1) If a bank offers to lend money to a company, the interest it charges is the yield that the bank wants to receive from investing in the company, because it can get just as good a return from lending the money to someone else. In other words, the interest rate is the opportunity cost of lending for the bank.

2) When shareholders invest in a company, the return that they can expect must be sufficient to persuade them not to sell some or all their shares and invest the money somewhere else. The yield on the shares is therefore the opportunity cost to the shareholders of not investing somewhere else.

Cost of equity has two meanings.

* Company - A Company's expected return on a prospective project or business opportunity.
* Investors - The cost of equity is the rate of return demanded by investors on their investments in a company.

A Company expects a return on projects undertaken or investment made. Investors demand a return on the funds invested in a Company. The amount of return is a percentage of the amount invested. This percentage is based upon the market rate of return for similar investments and the additional risks unique to the specific Company.

At least two methods can be used to calculate cost of equity.

i) Dividend valuation model (DVM)

ii) Capital Asset Pricing Model (CAPM)

Dividend Valuation Model

What gives a share value

The primary financial objective is assumed to be to maximise the wealth of equity shareholders.

If we view wealth as the value of the shares, it is necessary to have a theory as to what determines the value of an ordinary share.

The benefit to the shareholders from owning a share takes the form of future dividends and capital gain. It is the present value of these benefits that ~~can~~ creates the price to pay

price of share now (V_0) = present value of future dividends + present value of share price at terminal sale.

$$K_2 = \frac{D_0(1+g)}{V_2} + g$$

Where V_2 = Current market value of equity ex-div
 D_0 = Current dividend
 g = Growth rate in dividend
 K_2 = Cost of equity capital

Example.

The shares of Z plc are currently trading at \$24.20 per share. A dividend of \$4 per share was paid one year ago and dividends have been growing at 5% p.a. This growth in dividend is expected to continue. The company is expected to pay the next dividend next shortly. Calculate the cost of equity.

Solution.

$$V_2 = \$24.20 - \$4.20 = \underline{\underline{\$20}} \text{ ex-div}$$

$$g = 0.05$$

$$K_2 = \frac{\$4(\$1 + 0.05)}{\$20} + 0.05$$

$$K_2 = \underline{\underline{26\%}}$$

Practical Question.

A Company's share are currently valued at \$8.20 and the company is expected to pay an annual dividend of \$0.70 per share for the foreseeable future. The next annual dividend is payable in the future and the share price of \$8.20 is a cum dividend price.

Solution.

$$K_2 = \frac{D_0}{V_2} = \frac{\$0.70}{\$8.20 - \$0.70} = 0.093 \text{ or } 9.3\%$$

Estimating growth rate in earnings, where the value of growth rate is not given and there is enough data to enhance interpolation, then the growth rate should be computed. There are two methods of computing growth rate.

(I) Dividend growth valuation model / extrapolation of historical growth

(II) Gordon growth model / investment in earnings model.

(I) Dividend growth valuation method.

$$G = \sqrt[n]{\frac{\text{Value at end of period of } n \text{ years}}{\text{Value at start}}}$$

where n = number of time in series

G = Growth rate

(II) Gordon growth method / investment in earnings method.

where g =

where

g = growth rate

r = return on investment

b = retention ratio / proportion of earnings retained

Example.

A Company had provided the following dividend in recent years.

Year	Dividend
2001	100
2002	110
2003	120
2004	134
2005	148

Calculate the growth rate.

$$g = \sqrt[n]{\frac{\text{value at the end of period of } n \text{ years}}{\text{value at start}}} - 1$$

$$g = \sqrt[5]{\frac{148}{100}} - 1 = 0.103 = 10.3\%$$

on your calculator
press shift, $\sqrt{x}$ 1.48 - 1 = 0.103 = 10.3%.

Example 3.

You are given the following information about St Flour plc, which is an all equity financed company. The number of shares in issue is 150,000 of £1 each

Current Dividend £6158

Market value per share £83.42

Current earnings £62858

Net asset £315,000

Dividend 5 years ago £2473

You are required to estimate the cost of capital for the company.
 $\swarrow$ Dividend growth
 $\nwarrow$ Value in model.
 Gordon growth model.

First determine the growth rate.

$$\sqrt[5]{\frac{6158}{2473}} - 1 = 20\%$$

$$g = 20\%$$

$$k_e = \frac{D_0(1+g)}{MVE} + g$$

$$\frac{6158(1+0.20)}{150,000 \times £3.42} + 0.20$$

$$\frac{7389.6}{51300} + 0.20 = 0.2144$$

11.15%

Note: Where the question gives no. of years of dividend as in the above illustration then the \$ year is used direct (number of years) but where the dividend for the years are given year by year, then use $n-1$.

⑥

⑩ Using Gordon's growth model.

$$g = r \times b$$

Where g = growth rate

r = return on investment

b = retention rate / proportion of earnings retained.

$$r = \frac{62,858}{315,000} \times 100 = 20\%$$

$$b = \frac{62,858 - 6158}{62,858} = 90\%$$

$$g = 0.2 \times 0.9 = 18\%$$

$$k_e = \frac{D_0(1+g)}{MVE} + g$$

$$= \frac{6158(1+0.18)}{150,000 \times 1.342} + 18$$

$$k_e = \underline{\underline{19.4\%}}$$

CAPITAL ASSET PRICING MODEL (CAPM).

CAPM relates the expected return from an investment to the degree of risk from the investment. The required rate of return is made up of two parts.

i Risk Free return

ii Risk premium.

The risk premium is calculated by applying the project beta factor (β) to the difference

of the return.

$$R_S = R_F + \beta (R_M - R_F)$$

R_S = minimum required rate of return/cost of equity

R_F = Risk free rate of return

R_M = market portfolio's expected rate of return

$(R_M - R_F)$ = market premium for risk

$\beta (R_M - R_F)$ = Risk premium.

Illustration.

Calculate the cost of capital for Bdo plc from the data given below, using the Capital Asset Pricing model.

Beta coefficient for Bdo plc 0.5%

Expected rate of return on risk free securities 8%

Expected return on the market portfolio 12%

Solution

$$\begin{aligned} R_S &= R_F + \beta (R_M - R_F) \\ &= 8 + 0.5 (12 - 8) \\ &= 8 + 2 \\ &= \underline{\underline{10\%}} \end{aligned}$$

COST OF PREFERRED SHARE CAPITAL

This is the minimum rate of return required by preference shareholders in order to maintain their existing market price.

It is calculated as follows

$$k_{cp} = \frac{d}{mvp} \times 100$$

Where k_{cp} = Cost of preference share capital

d = latest dividend paid

mvp = market value of preference share

~~Value of redeemable debt~~

(8)

COST OF DEBT

Cost of irredeemable fixed rate debt or
Cost of irredeemable Debenture

This is the minimum rate of return required by debenture holders in order to maintain their existing market value

pre tax cost of
debt

post tax cost of
debt

~~where~~

$$rd = \frac{I}{MV}$$

where

$$rd = \frac{I(1-t)}{MV}$$

rd = the cost of debt capital

I = the annual interest rate

t = rate of tax on company profit

MV = Ex-interest market value of the debt

Note calculations are usually performed on a nominal value of 100 or 1000

Example cost of debt

The Coupon rate of interest on a Company irredeemable Bonds (perpetual Bonds) is 6% and the market value of Bonds is 103.60. The tax rate is 25%

1) Calculate the pre-tax cost of debt

2) Calculate the post tax cost of debt
solution

$$1) \text{ pre tax} = \frac{6}{103.60} = 0.058 \text{ or } 5.8\%$$

$$2) \text{ post tax} = \frac{6(1-0.25)}{103.60} = 0.043 \text{ or } 4.3\%$$

ABC plc has in issue 12% £500m redeemable debenture currently valued at £84. Interest is due for payment within the next two weeks. Corporation tax is 35%.

Calculate the cost of irredeemable debt.
Solution

$$\text{post tax rd} = \frac{1}{mv} (1 - t)$$

$$\frac{12\% (1 - 0.35)}{84 - 12}$$

$$\frac{12(65)}{72} \times 100 = 10.83\%$$

Cost of redeemable debt

This is calculated by the present value of future cash flows.

② To be received by the lender discounted at the pre-tax cost of debt, the lender requires rate of return

③ To be paid by the company (net of tax relief on the interest flows discounted at the post tax cost of debt).

Solution.

Year	Cash flows	DCF	
	Example	0.814%	100

A company has issued 7% loan stock. Annual interest has just being paid. The bond will be redeemed at par after four years.

The lender requires rate of return is 8.14%.

Required

Calculate the market value of the loan stock.

Year

	Cash Flows	Def @ 8.14%	PV
1 Interest	7.00	0.925	6.48
2 Interest	7.00	0.855	5.99
3 Interest	7.00	0.771	5.55
4 Interest	7.00	0.731	5.13
4 Redemption	100	0.731	73.10
			<u>96.25</u>

post tax cost of debt.

The current market value of a Company's 7% loan stock is 96.25. Annual interest has just been paid. The bond will be redeemed at par after four years. The rate of taxation of the Company's profit is 30%.

Required

Calculate the after tax cost of the bond for the Company.

Solution.

Current market value is 96.25 (treated as outflow),
annual interest net of tax = $100 \times 7\% \times (1 - 0.30) = 4.9$
Redemption value = 100.

Solution.

To determine the starting DCF, use the formula below.

$$ARR = \frac{1 + R - P}{n} \div \frac{R + P}{2}$$

$$\frac{1 + R - P}{n}$$

ARR = approximate IRR

$$100 \times 7\%$$

1 = annual interest net of tax

$$7 \times 1 - 0.3 = 4.9$$

R = Redemption

P = Current market price

n = number of years to redemption.

$$= \frac{4.9 + 100 - 96.25}{2} \div \frac{100 + 96.25}{2} = 0.0599$$

$$2$$

$$= 6\%$$

cash flow	Yr	FV	D.F. %	PV
0. initial value (96.25)	1.00	(96.25)	1.00	(96.25)
1 int. less tax 4.90	0.743	4.62	0.952	4.65
2 int less tax 4.90	0.290	4.36	0.907	4.44
3 int less tax 4.90	0.842	4.12	0.864	4.23
4 int less tax 4.90	0.792	3 3.88	0.823	4.03
4 Redemption 100	0.792	79.80	0.823	82.35

$$IRR = 5\% + \frac{3.41}{3.41 + 0.07} \times (6 - 5) = 5.98\% \Rightarrow \underline{6\%}$$

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Weighted Average Cost of Capital (WACC)

Method of Calculating WACC.

This known as Composite Cost of Capital, as it represents the cost of the various of financing in use. In calculating WACC, the company must first estimate the cost of each type of capital being used. The proportion of total capital coming from each source then weights the cost. The weighted costs are then added to give the overall cost of capital that is known as WACC.

Formula

$$WACC = r_E \frac{MVE}{MV_{Total}} + r_D \frac{MVD}{MV_{Total}} + r_P \frac{MP}{MV_{Total}}$$

Source of Finance

Market value $\times$ Cost (r)

Market value cost

r

Equity MVE

$\times$ RE

Preference Share MVE

$\times$ r-P

r-P-MVP

Debt

MVD

$\times$ r-D

r-D-MVD

Total

$\frac{\Sigma MV}{\Sigma r-MV}$

$\frac{\Sigma r-MV}{\Sigma r-MV}$

$$WACC = \Sigma r-MV / \Sigma MV$$

A company has 10 million shares each with a value of £4.20, whose cost is 7.5%. It has £30 million of 5% bonds with a market value of ~~101.00~~ and an after-tax cost of 3.5%.

It has a bank loan of £5 million whose after tax cost is 3.2%.

It also has 2 million 8% preference shares of £1 whose market price is £1.33 per share and and whose cost is 6%.

Calculate WACC.

Solution.

Source of Finance	Market Value	Cost	Market Value Cost
Equity	42.00	0.075	3.150
Preference Share	2.66	0.080	0.160
Bonds	30.80	0.035	1.061
Bank loan	5.00	0.032	0.160
	<u>79.96</u>		<u>4.531</u>

$$WACC = \frac{4.531}{79.96} = 0.05667 \text{ say } 5.7\%$$

Formula for WACC.

$$+ (0.075 \times 42/79.96) + (0.080 \times 2.66/79.96) + (0.035 \times 3.33/79.96) + (0.032 \times 5.0/79.96) = 0.05667 \text{ say } 5.7\%$$

